Coset residues and chamber orbits

Complete source disposition for L029

Human Compression project

This volume concerns D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162. The final journal edition is the fixed target.

The principal theorem is false for the full automorphism group in its stated definition of chirality. The accepted five-page counterexample is reproduced unchanged. The new synthesis supplies complete proofs of the valid coset and residue arguments, and records the smaller auxiliary corrections without conflating them with the principal error.

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Each manuscript retains its own pagination and bibliography. The three appended manuscripts retain their accepted bytes and proofs. All exact TeX inputs are supplied. A plain incidence diagram in the synthesis shows the seven-object auxiliary example.

General firmness in rank at least four, without an added action hypothesis, remains unproved. Its status is distinct from that of the source's false unrestricted statement. The finite checks certify their specific examples; the general proofs are written arguments. No new open-question solution, Lean result or compression factor is claimed.

The accompanying ledgers identify source ideas, retained project proofs, corrections, and borrowed background. The public principal-refutation record remains the record of the counterexample; this volume completes its surviving source obligations.

Coset residues and chamber orbits

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Abstract

We give a complete account of the valid residue arguments and the necessary corrections in Leemans–Tranchida's paper on residual connectedness. The rank-three result and a criterion using the specified group's action are proved. The published general criterion, which uses chirality under the full automorphism group, has a previously verified counterexample. Two smaller examples dispose of auxiliary assertions whose wording omits firmness or a rank restriction. The counterexamples, their scope and the corrected theorems are kept distinct. This manuscript assembles prior project proofs and the accepted corrections; it makes no additional conjecture or measurement claim.

1 Source and conventions

The source is the final journal edition of Leemans–Tranchida [1], pages 491–499. Its principal Theorem 1.1 asserts residual connectedness for the standard coset geometry of a C^+-group if that geometry is chiral. Chirality is defined using the *full* type-preserving automorphism group. That group need not be the specified C^+-group. The accepted five-page manuscript [4] gives an explicit counterexample with specified group $\mathrm{PSL}_2(13)$ and full group $\mathrm{PSL}_2(13) \times C_2$. It is reproduced in the complete volume with its proof unchanged.

We retain the source's definition of a geometry: every flag extends to a chamber. Firmness means that every panel has at least two completions; thinness means exactly two. Residual connectedness asks that every residue of rank at least two, including the whole geometry, have a connected incidence graph. None of these extra conditions is implicit in the word geometry. In particular, a rank-one residue with two objects has a disconnected incidence graph.

Let I be a finite type set. We use left cosets gH_i and incidence by nonempty intersection, retaining the type as part of an object. The source uses right cosets. Inversion sends $H_i g$ to $g^{-1}H_i$ and is an isomorphism of the two incidence systems. Thus this convention changes no flags, residues or transitivity assertions. Write $H_J = \bigcap_{j \in J} H_j$, with $H_\varnothing = G$. All general proofs below permit infinite groups.

The coset construction, the C^+ intersection condition and the canonical neighbouring chambers are prior results of the source and its cited predecessors. Sections 2–5 rework the existing project arguments [2, 3]. The chamber-component explanation in Section 6 is also retained from [3]. The principal counterexample [4], the rank boundary [5], and the seven-object example [6] have distinct reviewed proofs. Classical orbit arguments are not claimed as new. No new Lean proof is asserted by this assembly.

2 Flags and the internal residue map

The base objects H_i form a chamber. Each object is a translate of one of them. If gH_i and hH_j are incident, choose an element x of their intersection. Their pair is then $x\{H_i, H_j\}$. This proves the object and pair transitivity in the cited Tits construction, and also proves that every flag of rank at most two extends to a chamber.

Fix a proper subset $J \subset I$, and let $F_J = \{H_j : j \in J\}$. For $i \notin J$, put

$$S_{J,i} = \bigcap_{j \in J} H_j H_i.$$

For $J = \varnothing$ the intersection is G. The type-i objects of the residue of F_J are exactly the cosets gH_i with $g \in S_{J,i}$.

Lemma 2.1. *The maps*

$$h(H_J \cap H_i) \longmapsto hH_i \qquad (h \in H_J, i \notin J) \tag{1}$$

define a homomorphism from the internal coset incidence system $\Gamma(H_J; (H_J \cap H_i)_{i \notin J})$ *into the residue of* F_J*. It is injective on objects. It is surjective on objects if and only if* $S_{J,i} = H_J H_i$ *for every* $i \notin J$*.*

Proof. For $h, k \in H_J$, equality of hH_i and kH_i is equivalent to $k^{-1}h \in H_i$, and hence to $k^{-1}h \in H_J \cap H_i$. This proves both well-definedness and injectivity. A common element of two internal cosets is a common element of the corresponding external cosets, proving preservation of incidence. The image at type i consists of the cosets represented by $H_J H_i$. The full residue at that type is represented by $S_{J,i}$, and both sets are unions of left cosets of H_i. Equality is therefore exactly the surjectivity condition. □

A bijective incidence homomorphism need not be an isomorphism: its inverse may fail to preserve incidence. This distinction in the source's Remark 4.4 is essential. For a singleton $J = \{j\}$ the displayed surjectivity condition is automatic, but no assertion about the inverse follows from this alone.

Lemma 2.2. *Suppose the coset system is a geometry. The action of* G *is transitive on flags of type* $J \cup \{i\}$ *if and only if it is transitive on flags of type* J *and* $S_{J,i} = H_J H_i$*.*

Proof. Transitivity on the larger type implies transitivity on the smaller type, because every smaller flag has an extension in a geometry. For $g \in S_{J,i}$, the flag $F_J \cup \{gH_i\}$ is of the larger type. An element carrying the corresponding base flag to it fixes every member of F_J, so lies in H_J. Hence $gH_i = hH_i$ for some $h \in H_J$ and $g \in H_J H_i$. The reverse inclusion always holds.

Conversely, normalize the J-part of any flag of the larger type to F_J. Its last object is gH_i with $g \in S_{J,i}$. Write $g = hu$ with $h \in H_J$ and $u \in H_i$. Multiplication by h^{-1} fixes F_J and takes the last object to H_i. Thus every such flag is a translate of the base flag. □

This is the source's Lemma 4.5. If the maps (1) are surjective for every J, Lemma 2.2, by induction on the size of a flag, gives flag transitivity. In that case (1) is an isomorphism: any incident pair in a residue, together with F_J, is a flag and can be normalized by an element of H_J. That element is a common representative in the internal model. This proves the final implication in the cited source Lemma 4.3 for the maps used here.

Surjectivity for just one J does *not* supply transitivity on J. This is the missing condition in Proposition 4.6. The retained counterexample is $G = C_2^2$, with its three order-two subgroups H_0, H_1, H_2 and $H_3 = G$. Its geometry is the cone over the complete tripartite graph $K_{2,2,2}$, with seven objects and eight chambers. For $J = \{0, 1, 2\}$ the base residue and its internal coset model each have one object, so (1) is surjective. But a group of order four cannot be transitive on the eight J-flags. This is a counterexample to the stated equivalence, not to a C^+ assertion.

3 Connectivity and the specified action

Lemma 3.1. *Let K_i be subgroups of a group K and let the index set have at least two elements. The incidence graph of their coset system is connected if and only if $K = \langle K_i : i \in I \rangle$.*

Proof. The base cosets form a clique. Each K_i fixes one of its vertices, and therefore preserves the connected component of that clique. If the subgroups generate K, every translate of every base vertex lies in this component, proving connectedness. Conversely, a path starting at a base vertex can be followed by successively choosing elements in the intersections of its consecutive cosets. Each step changes the representative by multiplication by an element of one of the K_i. Every vertex reached therefore has a representative in $\langle K_i : i \in I \rangle$. If all vertices are reached, each coset of any one K_i has such a representative, so their union gives $K = \langle K_i : i \in I \rangle$. □

Proposition 3.2. *Suppose G is transitive on flags of every proper type in a coset geometry. If $|I \setminus J| \geq 2$ and*

$$H_J = \langle H_J \cap H_i : i \in I \setminus J \rangle, \tag{2}$$

then every residue of a flag of type J is connected.

Proof. Normalize the flag to F_J. For every $i \notin J$, the set $J \cup \{i\}$ is still proper, so Lemma 2.2 and Lemma 2.1 make the internal model bijective on objects. Equation (2) and Lemma 3.1 make that internal graph connected. Its image is a connected spanning subgraph of the actual residue. Any extra incidence edges preserve connectedness. □

This is the sufficient residue statement used in the source, with its action and rank hypotheses explicit. In rank one, transitivity on the enlarged type would mean transitivity on chambers, which has not been assumed. The rank-one boundary of its printed Proposition 4.7 is exhibited in [6]. The source's preceding paragraph already excludes these residues from the residual-connectedness question.

4 Canonical chambers of a marked group

From now on let $|I| = r \geq 3$, with a distinguished element 0. Let G be generated by distinct nonidentity elements α_i for $i \neq 0$, and put $\alpha_0 = 1$. Set

$$a_{ij} = \alpha_i^{-1}\alpha_j, \qquad P_S = \langle a_{ij} : i, j \in S \rangle, \qquad H_i = P_{I \setminus \{i\}}.$$

Assume the C^+ intersection condition

$$P_S \cap P_T = P_{S \cap T} \qquad (|S|, |T| \geq 2). \tag{3}$$

For an empty or singleton set S, the group P_S is trivial, so (3) holds also when either index set has size less than two. Repeated intersection therefore gives

$$H_J = P_{I \setminus J}. \tag{4}$$

For $i > 0$, H_i is generated by the α_j with $j \neq i$; for $i = 0$, it is generated by $\alpha_1^{-1}\alpha_j$ for $j \geq 2$. Thus these are precisely the maximal parabolics in Construction 3.2.

Lemma 4.1. *Let $C = \{H_i : i \in I\}$. For $i \in I$ and any $j \neq i$, put*

$$D_i = (C \setminus \{H_i\}) \cup \{a_{ij}H_i\}.$$

This does not depend on j. It is a chamber distinct from C, adjacent to it at type i. Moreover

$$a_{ji}D_i = D_j, \qquad D_i = \alpha_i^{-1}D_0, \qquad D_0 \text{ is } i\text{-adjacent to } \alpha_i C. \tag{5}$$

The two G-orbits GC and GD_0 are disjoint.

Proof. For $j, k \neq i$, the quotient $a_{ij}^{-1} a_{ik} = a_{jk}$ belongs to H_i, proving independence of the choice. The coset differs from H_i because $a_{ij} \neq 1$ and $P_{\{i,j\}} \cap H_i = P_{\{j\}} = 1$. If $\ell \neq i, j$, its representative a_{ij} belongs to H_ℓ. To obtain incidence with H_j, choose $k \neq i, j$; this is where $r \geq 3$ is needed. The element $a_{ik} = a_{ij} a_{jk}$ lies in both $a_{ij} H_i$ and H_j. All unchanged objects meet at the identity, so D_i is a chamber.

For the first identity in (5), at type i the two multipliers cancel; at type j the multiplier produces the altered coset; at every other type it lies in that parabolic. Taking one index to be zero gives the second identity. Translating the adjacency of C, D_i by α_i gives the last assertion in (5). If $gC = D_0$, then g fixes the base panel missing type zero. Its stabilizer is $H_{I \setminus \{0\}} = P_{\{0\}} = 1$ by (4). This would give $D_0 = C$, a contradiction. □

The lemma is the valid rank-at-least-three form of source Lemma 4.8. It describes neighbours of these canonical chambers. It does not normalize an arbitrary chamber to C.

Proposition 4.2. *If the marked coset system is a geometry and its full type-preserving group A has two alternating chamber orbits, then it is thin. For an arbitrary firm geometry with such a group action, A is transitive on flags of every proper type, and residual connectedness can be tested on the residues of a single chamber.*

Proof. In the marked case, C and D_0 are adjacent and therefore lie in the two different A-orbits. Lemma 4.1 supplies a neighbour of every type for both orbit representatives. Transport by A supplies one for every chamber. Every panel has at least two completions. It cannot have three: three chambers completing one panel are pairwise adjacent, and three pairwise adjacent chambers cannot lie alternately in just two orbits. Thus the geometry is thin.

For the general firm case, let F be a flag of proper type and choose a chamber containing it. At a type missing from F, firmness gives an adjacent chamber still containing F. These two chambers lie in the two different orbits. Given another flag of the same type, extend it to a chamber and compare with the first or second chamber as required. The resulting automorphism maps the whole typed flag to the other one. Thus A is transitive on each proper flag type, and their residues are isomorphic. Every proper type occurs in a chosen chamber; this proves the residue test, including all residues of rank at least two. □

This proves the useful firm versions of source Theorem 4.1 and Proposition 4.2. It does not prove their unrestricted wording: the seven-object example in [6] satisfies the source's literal chirality definition and refutes both. Nor does this proposition give proper-flag transitivity for the subgroup G.

5 The valid residual-connectedness criterion

Theorem 5.1. *For the nondegenerate C^+ marking above, suppose its coset system is a geometry and G is transitive on flags of every proper type. Then the geometry is firm and residually connected.*

Proof. Each panel can be normalized to a base panel. Its two canonical completions from Lemma 4.1 give firmness. Let F_J be a base flag, put $S = I \setminus J$, and assume $|S| \geq 2$. As in Proposition 3.2, the internal residue model is bijective on objects and preserves incidence.

If $|S| \geq 3$, equation (4) identifies its ambient group with P_S and its parabolics with $P_{S \setminus \{i\}}$ for $i \in S$. Every generator a_{uv} of P_S belongs to one of those parabolics: choose $i \in S \setminus \{u, v\}$. They therefore generate P_S, and Proposition 3.2 proves connectedness.

If $S = \{u, v\}$, write $a = a_{uv}$, so $H_J = \langle a \rangle$. Its two internal parabolics are trivial. The residue has two copies of this cyclic group as its objects. Translates of C give edges from $a^k H_u$ to $a^k H_v$. Translates of D_v give edges from $a^k H_u$ to $a^{k-1} H_v$. Here $a \neq 1$, and distinct powers

represent distinct cosets until the order of a is reached, by (4). These two matchings form a cycle of length $2|a|$ if a has finite order, and a doubly infinite path otherwise. Their union is connected and spans the residue. Possible additional edges cannot destroy connectedness. This also explains the order-three diagram of source Example 4.10; reversing both cyclic coordinates changes the sign of the shift without changing the argument.

Finally every flag of proper type can be normalized to F_J, so the conclusion holds for all residues, including the empty flag's residue. □

Corollary 5.2. *Every rank-three system from a nondegenerate C^+ marking is a firm, residually connected geometry.*

Proof. Every incident pair is a translate of a base pair and extends to a chamber. Objects and the empty flag extend too; any flag of size three is already a chamber. Thus it is a geometry. Every proper flag has rank at most two, so the coset action is transitive on its type. Theorem 5.1 applies. □

This is source Proposition 4.11. It requires neither chirality nor a full automorphism computation.

For rank two, the nontrivial cyclic marking has both parabolics trivial and gives disjoint edges. The full written proof and literal check in [5] refute the unrestricted Lemma 4.8 and Corollary 4.9. Imposing rank at least four in a proposed replacement for Corollary 4.9 produces a different assertion, unproved here. A false universal statement is not recovered by silently changing its hypotheses, and its counterexample does not automatically create a new source obligation for every such replacement.

6 Why the two acting groups differ

The following retained argument pinpoints the principal error. It adds a structural explanation to the explicit counterexample.

Theorem 6.1. *If the marked coset system is a thin geometry, then $\Omega = GC \dot\cup GD_0$ is one gallery component. It contains every object and every incident pair. The stabilizer of C in the full type-preserving group A is trivial. If A is chiral, then $\mathrm{Stab}_A(\Omega) = G$ and the gallery components are indexed by A/G.*

Proof. Thinness makes the i-neighbour of gC equal to gD_i, which lies in GD_0. The i-neighbour of gD_0 is $g\alpha_i C$. Thus Ω is closed under adjacency. The two-step galleries $C, D_0, \alpha_i C$, their reversals, and their translates join C to every gC, because the α_i generate G. Each gD_0 is adjacent to gC. Consequently Ω is a connected component.

Every object occurs in some gC. For an incident pair, an element of the intersection of the two cosets is a common representative, placing the pair in a translate of C. An automorphism fixing C fixes its neighbours, since each panel has a unique other completion. Repetition along galleries fixes Ω pointwise. Since Ω contains every object, the automorphism is the identity. The coset action itself is faithful because its kernel fixes C and lies in $\bigcap_i H_i = P_\varnothing = 1$.

Suppose now A is chiral. If $f \in A$ preserves Ω, then fC lies in GC or GD_0. The second possibility would identify the two A-orbits of the adjacent chambers C, D_0. Hence $fC = gC$ for some $g \in G$. The element $g^{-1}f$ fixes C and is the identity, so $f = g$. Conversely G preserves Ω. Every chamber is an A-translate of C or D_0, so every component is an A-translate of Ω. The orbit-stabilizer bijection gives A/G. □

The theorem does not assert that A/G has one element. The accepted example [4] has $|G| = 1092$, $|A| = 2184$, two gallery components, and a disconnected rank-two residue. Its full-group chirality supplies the hypotheses of the published Theorem 1.1 and disproves its

conclusion. The switch from proper-flag transitivity for A to that property for G is therefore a substantive error, separate from the smaller missing rank and firmness conditions.

7 Disposition of the source

Figure 1 illustrates the smaller counterexample from [6]. Its chambers are the four triangles $C_j = \{c_j, x_j, y_j\}$ and $D_j = \{d, x_j, y_j\}$, for $j = 1, 2$. The full typed group exchanges the two halves. Its chamber orbits are $\{C_1, C_2\}$ and $\{D_1, D_2\}$, and the only adjacencies join C_j to D_j. Yet the panel $\{c_j, x_j\}$ has only one completion. This is the missing-firmness issue; it is separate from the principal $\mathrm{PSL}_2(13)$ counterexample.

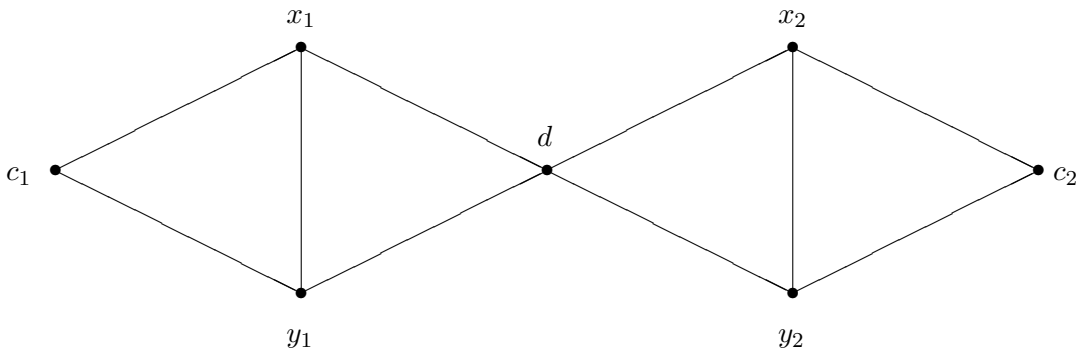


Figure 1: All incidences of the seven-object example. The type classes are $\{d, c_1, c_2\}$, $\{x_1, x_2\}$ and $\{y_1, y_2\}$.

The complete source ledger accompanies this manuscript. The following table records the mathematical outcomes; it is not a claim that false statements have been reproved.

Source assertion	**Disposition and proof**
Theorem 1.1	Refuted in its literal full-group scope by the accepted five-page counterexample [4]. The valid specified-action criterion is Theorem 5.1.
Proposition 2.1	Cited Tits construction; object and pair normalization proved in Section 2.
Theorem 2.2	Cited flag-transitive residue theorem from the source's reference [1]. Its sufficient part needed here is proved in Section 3; the whole cited monograph is not an additional intake.
Theorems 3.1, 3.3; Corollary 3.4	Explicitly borrowed C-group and C^+ correspondence results from the source's reference [6], *Highly symmetric hypertopes*. Their attribution is retained; the present corrections do not refute them.
Construction 3.2	The parabolics are identified with $P_{I\setminus\{i\}}$ in Section 4; left and right cosets are related by the explicit inversion map in Section 1.
Chirality implies thinness; Theorem 4.1; Proposition 4.2	False without firmness under the stated geometry definition; the seven-object example [6] supplies the full group and all residues explicitly. The firm normalization statements are proved in Proposition 4.2.
Lemma 4.3; Remark 4.4	The cited internal map, its object-surjectivity criterion, and its distinction from an isomorphism are proved in Section 2 for the interfaces used here.

Lemma 4.5	Recovered as Lemma 2.2, with the smaller flag-type transitivity present.
Proposition 4.6	Refuted by the retained C_2^2 example in Section 2. Adding smaller-type transitivity repairs the equivalence.
Proposition 4.7	The displayed rank-one boundary is false, as shown in [6]; the preceding source paragraph focuses on rank at least two. Proposition 3.2 states the valid rank and action hypotheses.
Lemma 4.8; Corollary 4.9	Refuted in unrestricted rank by [5]. The canonical-neighbour lemma for rank at least three is Lemma 4.1. General firmness in rank at least four without further action hypotheses remains unproved.
Example 4.10; Figure 1	The cyclic two-matching construction is proved for all cyclic orders in Theorem 5.1; the source's order-three illustration is a special case.
Proposition 4.11	Recovered as Corollary 5.2.

No named open question is solved or added by this disposition. The principal refutation, valid source results, missing hypotheses, and useful unproved variants are separate outcomes. Existing computational certificates are reused without repeating their acquisition. The exact finite checks in [5, 6] verify their stated examples, not the arbitrary-group proofs above. Compression measurements and further Lean formalization are deferred.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162.

[2] Human Compression project, *Incidence results derived from relation matrices*, retained proof bank, Section 1.

[3] Human Compression project, *Chamber components in marked coset geometries*, retained phase276 manuscript, Sections 1–5.

[4] Human Compression project, *A chiral coset geometry with disconnected residues*, accepted five-page manuscript and exact certificate.

[5] Human Compression project, *Panels in the coset construction for a C^+-group*, accepted three-page manuscript, with the rank boundary and the retained finite panel checks.

[6] Human Compression project, *Chirality without firmness*, accepted two-page manuscript and seven-object certificate.

A chiral coset geometry with disconnected residues

Human Compression project

Abstract

We give a rank-four counterexample to the stated residual-connectedness criterion for chiral geometries constructed from C^+-groups. The specified group is $\mathrm{PSL}_2(13)$, whereas the full type-preserving automorphism group is $\mathrm{PSL}_2(13) \times C_2$. The geometry has 112 objects and 4368 chambers. Its full group has two chamber orbits, with adjacent chambers in different orbits, but a rank-two residue is the disjoint union of two six-cycles. We prove the full automorphism assertion by an elementary calculation on thirteen points; no automorphism-group software or classification is required.

1 The decorated point graph

All entries in this note are in $\mathbb{F} = \mathbb{F}_{13}$. Let

$$K = \{1, 3, 4, 9, 10, 12\} = (\mathbb{F}^\times)^2, \qquad V = (\mathbb{F}^2 \setminus \{0\})/K.$$

Write $[v]$ for the class Kv. There are 28 classes, two on each projective line. Define a simple undirected graph X on V by

$$[v] \sim [w] \quad \Longleftrightarrow \quad \det(v, w) \in K. \tag{1}$$

The condition is independent of the representatives and symmetric because $-1 \in K$. Points on the same projective line are not adjacent. Put $x = [(1, 0)^T]$, $y = [(0, 1)^T]$, and $z = [(1, 1)^T]$.

The natural $\mathrm{SL}_2(13)$ action preserves (1). Its kernel is $\{\pm I\}$: a matrix fixing x, y, z must be scalar, and a determinant-one scalar has square one. Hence its image $G = \mathrm{PSL}_2(13)$ has order $13(13^2-1)/2 = 1092$. It is transitive on vertices and on ordered edges. Indeed a nonzero vector can be completed to a determinant-one basis; for an adjacent ordered pair one can first rescale one vector by a member of K to obtain the required determinant. Every vertex has thirteen neighbours: those of x have unique representatives $(a, 1)^T$, $a \in \mathbb{F}$. The graph induced on this neighbourhood is

$$a \sim b \quad \Longleftrightarrow \quad a - b \in K, \tag{2}$$

the Paley graph on thirteen points. In particular every edge of X has six common neighbours.

Multiplication of every vector by 2 induces an automorphism s of X, since $4 \in K$. It is an involution and commutes with G. It does not belong to G. Otherwise a determinant-one matrix inducing this action would be $2kI$ for some $k \in K$, forcing $2k = \pm 1$, whereas $\pm 7 \notin K$. Thus

$$A = \langle G, s \rangle \cong G \times C_2, \qquad |A| = 2184. \tag{3}$$

Every element of A is induced by an invertible linear matrix, taken modulo the scalar subgroup K.

Lemma 1. *The full automorphism group of X is A.*

Proof. The other decorated point on the line of $[v]$ is $[2v]$. For each different line, exactly one of its two decorations is adjacent to $[v]$, and the other is adjacent to $[2v]$. These two vertices therefore have no common neighbour. Two vertices on distinct lines do have a common neighbour: solve the independent linear equations $\det(v, w) = \det(v', w) = 1$. Thus $[2v]$ is the unique other vertex with no common neighbour with $[v]$, and every automorphism preserves this antipodal pairing.

An automorphism fixing x and its thirteen neighbours fixes their antipodes and the antipode of x, hence every vertex. Consequently the stabilizer of x embeds in the automorphism group of (2). We bound the latter group directly. The neighbours of 0 in (2) induce the six-cycle, in the following cyclic order:

$$(1, 4, 3, 12, 9, 10). \tag{4}$$

Number its positions $0, \ldots, 5$. The six remaining nonzero vertices have the following neighbour sets on this cycle:

vertex	2	5	6	7	8	11
positions	023	014	245	125	134	035.

(5)

The six triples in (5) are the translates of $\{0, 2, 3\}$ in $\mathbb{Z}/6\mathbb{Z}$. They are distinct, so an automorphism fixing 0 is determined by its action on (4). This action belongs to the dihedral group of the cycle. Every rotation preserves the collection (5), while the reflection $i \mapsto -i$ sends $\{0, 2, 3\}$ to $\{0, 3, 4\}$, which is absent. No reflection preserves the collection. The stabilizer of 0 in the automorphism group of (2) therefore has order at most six, and that group has order at most $13 \cdot 6 = 78$.

It follows that $|\operatorname{Aut}(X)| \leq 28 \cdot 78 = 2184$. The subgroup (3) attains the bound, proving equality. □

There are $28 \cdot 13 \cdot 6 = 2184$ ordered triangles in X. An element of A fixing one of them fixes three different projective lines and is scalar; fixing the decorated points then makes it the identity modulo K. Thus A acts freely and transitively on the ordered triangles. In particular each triangle has the same number of common neighbours as x, y, z. A common neighbour of x, y is represented by $(a, 1)^T$ with $a \in K$. It is adjacent to z precisely when $1 - a \in K$. Directly from the six displayed elements of K, the solutions are $a = 4, 10$. Every triangle has exactly two common neighbours.

2 The geometry and its full chirality

Let $I = \{0, 1, 2, 3\}$. Take four tagged copies of V, and declare objects of different types incident precisely when their underlying vertices are adjacent in X. Equality is the only incidence within one type. Denote this incidence system by Γ. Every incident pair extends to a triangle; every triangle has exactly two common neighbours. Hence every flag extends to a chamber and every panel has exactly two completions. Thus Γ is a thin geometry. Its objects number $4 \cdot 28 = 112$, and its chambers number $2184 \cdot 2 = 4368$.

Lemma 2. *The full type-preserving automorphism group of* Γ *is the diagonal action of* A*. It has exactly two chamber orbits, and adjacent chambers belong to different orbits.*

Proof. First, distinct vertices of X have distinct neighbourhoods. Antipodes have disjoint neighbourhoods. If u, v are on different lines, one of the two decorations on the line of u is adjacent to v, while neither is adjacent to u; their neighbourhoods therefore differ.

Write a type-preserving automorphism as permutations f_i on the four copies of V. For distinct i, j, incidence gives $N(f_i(v)) = f_j(N(v))$. Given i, k, choose j distinct from both. The preceding equality gives $N(f_i(v)) = N(f_k(v))$, so the absence of equal neighbourhoods

implies $f_i = f_k$. All four permutations are the same automorphism of X. Lemma 1 proves the first assertion.

This group is free on chambers, since it is free on ordered triangles. Its order and the chamber count give exactly two orbits. If an automorphism took one chamber to an adjacent chamber, it would fix the three common objects and hence be the identity. Adjacent chambers are distinct, so this cannot happen. This is chirality under the full type-preserving group, not merely under a proposed subgroup. □

The geometry is not residually connected. Fix the type-1 object x and the type-2 object y. The remaining objects in their residue are two copies, of types 0 and 3, of

$$\{[(a,1)^T] : a \in K\}.$$

Their incidence is $a - b \in K$, so this residue is the bipartite double of the cycle (4). Since that cycle has even length, its bipartite double is two disjoint six-cycles: one component uses the even cycle positions in type 0 and the odd positions in type 3, and the other uses the opposite positions.

3 The exact C^+ coset construction

It remains to verify the algebraic hypothesis in the source theorem. In $G = \mathrm{PSL}_2(13)$ use the determinant-one matrices

$$\alpha_0 = I, \qquad \alpha_1 = \begin{pmatrix} 4 & 0 \\ 7 & 10 \end{pmatrix}, \quad \alpha_2 = \begin{pmatrix} 4 & 6 \\ 0 & 10 \end{pmatrix}, \quad \alpha_3 = \begin{pmatrix} 3 & 0 \\ 0 & 9 \end{pmatrix}. \tag{6}$$

Choose the following base objects, in type order:

$$v_0 = [(4,1)^T], \qquad v_1 = x, \qquad v_2 = y, \qquad v_3 = z. \tag{7}$$

They form a chamber, as their six determinants belong to K. For $S \subseteq I$ put $P_S = \langle \alpha_i^{-1}\alpha_j : i,j \in S \rangle$. The six differences for $i < j$ are as follows; the last column gives their two fixed base objects.

ij	$\alpha_i^{-1}\alpha_j$	fixed labels
01	$\left(\begin{smallmatrix} 4 & 0 \\ 7 & 10 \end{smallmatrix}\right)$	$2,3$
02	$\left(\begin{smallmatrix} 4 & 6 \\ 0 & 10 \end{smallmatrix}\right)$	$1,3$
03	$\left(\begin{smallmatrix} 3 & 0 \\ 0 & 9 \end{smallmatrix}\right)$	$1,2$
12	$\left(\begin{smallmatrix} 1 & 8 \\ 11 & 11 \end{smallmatrix}\right)$	$0,3$
13	$\left(\begin{smallmatrix} 4 & 0 \\ 5 & 10 \end{smallmatrix}\right)$	$0,2$
23	$\left(\begin{smallmatrix} 4 & 11 \\ 0 & 10 \end{smallmatrix}\right)$	$0,1$

(8)

Each is non-scalar and its cube is I or $-I$, so it has order three in G. Each fixes the two displayed decorated points, not just their projective lines. No nonidentity projective transformation fixes three distinct lines.

The stabilizer in G of any decorated point is $C_{13} : C_3$, of order 39. For x this follows directly from its upper triangular lifts, whose first diagonal entry belongs to K; the others follow by transitivity. For each i, (8) puts $P_{I\setminus\{i\}}$ in this stabilizer of v_i and supplies two distinct subgroups of order three, distinguished by their second fixed lines. A subgroup of $C_{13} : C_3$ containing two such subgroups must be the whole group: if it met the normal C_{13} trivially it would embed in C_3, and a nontrivial intersection is the whole C_{13}. Thus

$$H_i := P_{I\setminus\{i\}} = G_{v_i}, \qquad |H_i| = 39. \tag{9}$$

The three matrices generate G. Indeed H_1 contains every upper unipotent matrix and H_2 every lower unipotent matrix. These generate $\mathrm{SL}_2(13)$ before passing to its projective quotient. For instance the explicit products

$$\alpha_2\alpha_3 = \begin{pmatrix} -1 & 2 \\ 0 & -1 \end{pmatrix}, \qquad \alpha_1\alpha_3 = \begin{pmatrix} -1 & 0 \\ 8 & -1 \end{pmatrix}$$

already give nonzero upper and lower translations, whose powers give all such translations. Row elimination then gives every determinant-one matrix.

The intersection property is now explicit. Two distinct groups in (9) intersect in a group of order three, since G is transitive on the $28 \cdot 13$ ordered edges. Their intersection is the group P_S from (8) on their two complementary labels. Three distinct groups in (9) intersect trivially. A cyclic P_S with $|S| = 2$ is contained in H_i exactly when $i \notin S$; otherwise an element of the intersection would fix three base lines. Two different such cyclic groups likewise intersect trivially. Together with $P_I = G$ and $P_S = 1$ for $|S| \leq 1$, these observations prove

$$P_S \cap P_T = P_{S \cap T} \qquad (S, T \subseteq I). \tag{10}$$

The generators are independent: omitting α_i, $i > 0$, leaves H_i of order 39. Hence (6) is a C^+-marking in the source's exact convention.

Finally the source's right coset $H_i g$ corresponds to $g^{-1}v_i$. Equality of fibres follows from (9). Coset incidence is precisely the G-orbit of the corresponding base pair; every base pair in (7) is an edge, and G is transitive on all ordered edges. Therefore all six coset incidence relations agree with (1). This proves that Γ is exactly Construction 3.2 for (6), rather than a geometry which merely contains a suitable group action.

Theorem 3. *The C^+-group (6) has a chiral coset geometry that is not residually connected. Its full type-preserving group is $\mathrm{PSL}_2(13) \times C_2$, its specified group is $\mathrm{PSL}_2(13)$, and its rank-two residue at the type-1 and type-2 objects of (7) is two disjoint six-cycles.*

4 Source scope and verification

The theorem contradicts Theorem 1.1 in the fixed published edition of Leemans–Tranchida [1]. Their definition on printed page 493 uses the full $\mathrm{Aut}_I(\Gamma)$ to define chirality; Lemma 2 verifies exactly that definition. Construction 3.2 on page 494 is paid by (6)–(10) and the coset identification. The geometry is thin and every flag extends, so neither of these hypotheses is missing.

The issue is the passage on printed page 496 from proper-flag transitivity under the full group to transitivity under the specified G. Here the full group acts freely and transitively on flags of any three fixed types, while G has two orbits on those flags. The internal rank-two coset model thus covers only one of the two actual residue cycles. This counterexample does not refute the separately valid rank-three conclusion in Proposition 4.11, nor a criterion which explicitly requires the specified group to have the two alternating chamber orbits. It is a correction of a published theorem, not a claim to have solved a named open conjecture.

The decorated-vector vocabulary comes from the accompanying recovery of the tetrahedral construction of Catalano, Fernandes, Hubard and Leemans [2]. That source uses point and tetrahedron types. The present construction instead takes four point types. All group and incidence arguments needed for this example are given here. We make no claim of priority for the graph (1), the Paley graph, or their standard matrix actions.

The accompanying short program independently enumerates the 28 vertices, all triangles and four-cliques, the generated 1092-element matrix group and all 256 intersections in (10). It checks the full ordered-triangle action, both chamber orbits, the six local attachment triples and the disconnected actual residue. These are finite checks of the displayed example; the full automorphism upper bound is the written argument of Lemma 1.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics 4 (2021), 491–499, doi:10.5802/alco.162.

[2] D. Catalano, M. E. Fernandes, I. Hubard and D. Leemans, *Hypertopes with tetrahedral diagram*, Electronic Journal of Combinatorics 25(3) (2018), P3.22.

Panels in the coset construction for a C^+-group

Human Compression project

Abstract

We distinguish a rank restriction missing from the literal statement of Corollary 4.9 of Leemans–Tranchida from the unresolved assertion in higher rank. We also give an exact criterion for geometry and firmness in rank four, using products of the four maximal parabolics. This reduces a possible finite counterexample to a small certificate without an automorphism computation. The criterion is checked on four previously acquired markings; none is a counterexample in rank four.

1 The rank restriction

We use the final published edition of [1], printed pages 493–494 and 497. Put $\alpha_0 = 1$ and

$$P_J = \langle \alpha_i^{-1}\alpha_j : i, j \in J \rangle.$$

The C^+ intersection condition is $P_J \cap P_K = P_{J\cap K}$ for $|J|, |K| \geq 2$. Construction 3.2 takes $H_i = P_{I\setminus\{i\}}$ and forms the incidence system of right cosets $H_i g$, with incidence defined by nonempty intersection. A geometry means that every flag extends to a flag containing one object of each type. Connectedness is a separate condition.

Proposition 1. *For every nontrivial cyclic group $G = \langle a \rangle$, the rank-two marking $R = \{a\}$ satisfies the stated C^+ condition and is independent. Its associated coset system is a geometry, but is not firm.*

Proof. Here $I = \{0, 1\}$, $\alpha_0 = 1$, and $\alpha_1 = a$. The only subset of I with at least two elements is I itself, so the required intersection equality is $G \cap G = G$. Independence says $a \notin \langle \varnothing \rangle = 1$, which holds. Both maximal parabolics are trivial, since their defining index sets are singletons. Each type therefore has one object for each $g \in G$. The type-zero object labelled g is incident to the type-one object labelled h exactly when $g = h$. The geometry consists of disjoint edges. Every flag extends to an edge, but every one-object panel has exactly one completion. This is the failure of firmness. □

No lower rank bound is stated in the displayed C^+ definition, Construction 3.2 or Corollary 4.9. The proof of Lemma 4.8, however, chooses an index k distinct from two prescribed indices i, j. Consequently this example identifies a missing rank restriction in the literal wording of both Lemma 4.8 and Corollary 4.9: replacing one face of a matching edge gives a nonincident pair, not an adjacent chamber. Section 2.1, printed page 492, imposes neither connectedness nor a lower rank bound on a geometry, and its Proposition 2.1 permits every positive number of types. The later definitions do not add a global restriction. The counterexample does not decide the intended statement for rank at least three. In rank three the desired conclusion follows from pair transitivity and the canonical adjacent chambers, as in the existing project proof of Proposition 4.11. For rank at least four, the remaining problem is to supply a neighbour for a panel outside the orbit of a canonical panel. The accepted counterexample to the principal residual-connectedness theorem is thin, and does not solve this problem.

2 A criterion in rank four

The next reduction applies to arbitrary subgroups of a finite group; the C^+ condition is needed only for the last description of the stabilizer. In particular, no transitivity on triple flags is assumed.

Lemma 2. *Let G be finite, let $I = \{0,1,2,3\}$, and let $H_i \le G$. For each missing type d, choose an ordering (a,b,c) of $I \setminus \{d\}$ and put*

$$X_c = H_cH_a \cap H_cH_b, \qquad Y_d(x) = H_dH_a \cap H_dH_b \cap H_dH_cx \quad (x \in X_c). \tag{1}$$

The coset system is a geometry if and only if $Y_d(x) \neq \varnothing$ for all four choices of d and all $x \in X_c$. It is a firm geometry if and only if

$$|Y_d(x)| \ge 2|H_d| \quad \text{for all these } d, x. \tag{2}$$

It suffices in either test to take one representative for each orbit of $K = H_a \cap H_b$ on the right cosets H_cx contained in X_c. For a rank-four C^+ marking, $K = P_{\{c,d\}} = \langle \alpha_c^{-1}\alpha_d \rangle$ is cyclic.

Proof. Every incident pair can be taken to H_a, H_b by right multiplication: choose an element z in the intersection of its two cosets and multiply both cosets by z^{-1}. Every pair, every object and the empty flag therefore extend to a canonical chamber after the same normalization. The only possible obstruction to being a geometry is a triple flag with no completion at the fourth type.

After normalizing the types a, b of such a triple, its last object is H_cx. It intersects both H_a and H_b precisely when $x \in X_c$. A type-d coset H_dy completes the triple precisely when y belongs to each of the three sets defining $Y_d(x)$. Each such set, and hence their intersection, is a union of right cosets of H_d. The number of completions is therefore exactly $|Y_d(x)|/|H_d|$. This proves the geometry criterion and (2). Once all panels have at least two completions, every flag of corank one has a residue with at least two objects, which is precisely firmness.

Changing x to hx with $h \in H_c$ leaves $Y_d(x)$ unchanged. Right multiplication by $k \in K$ fixes H_a, H_b and sends the panel with third object H_cx to the panel with third object H_cxk. It bijects their completions; equivalently $Y_d(xk) = Y_d(x)k$. Thus one representative from each stated orbit suffices. Finally the intersection condition gives $H_a \cap H_b = P_{I\setminus\{a,b\}}$. The subgroup for a two-element index set is generated by either one of the mutually inverse differences, giving the cyclic description. □

3 Finite checks and remaining scope

The accompanying program checks the rank-two example for $G = C_2$ by enumerating all flags: there are four objects, two chambers, and four panels, each with one completion. This check is separate from the uniform written proof for all nontrivial cyclic groups.

For rank four the program reuses the four stored $\mathrm{PSL}(2,7)$ markings from the earlier diagnostic. It checks the multiplication table against determinant-one matrices modulo their central signs, regenerates all sixteen index subgroups, and verifies their 256 intersection equalities. It checks (1) against literal coset intersections for every normalized panel, then partitions the panels into the stated stabilizer orbits. No marking is searched for anew. Each of the four geometries has five completions at every panel. The existing projective-line description gives the same conclusion: four typed copies of an eight-point set are incident by inequality.

For a future fixed finite group, a proposed counterexample must pass the nondegeneracy and full intersection checks, have no zero entry in (1), and have at least one entry equal to

$|H_d|$. A singleton panel without the first of these two completion requirements would not refute Corollary 4.9: the coset system might fail to be a geometry. This packet does not produce a counterexample or a general proof in rank at least four, and it gives no whole-original, conjecture, formalization or compression credit.

The C^+ construction and canonical neighbour formula are due to the authors of [1] and their cited predecessors. The previously written chamber-component argument and the four stored markings are reused. The present additions are the explicit rank boundary and the panel-counting reduction; no claim of historical novelty for the elementary coset calculation is made.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162.

Chirality without firmness

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In the final published edition of Leemans–Tranchida [1], printed page 492, a geometry is an incidence system in which every flag extends to a chamber. Firmness is defined separately. On page 493, chirality requires exactly two orbits of the full type-preserving group on chambers, with adjacent chambers in different orbits. These definitions permit singleton panels. The following example shows why the omission matters for the unrestricted auxiliary statements.

1 An example with seven objects

Let the three type classes be

$$X_0 = \{d, c_1, c_2\}, \qquad X_1 = \{x_1, x_2\}, \qquad X_2 = \{y_1, y_2\}.$$

Apart from reflexivity, declare precisely the following pairs incident, for $j = 1, 2$:

$$x_j \sim y_j, \qquad d \sim x_j, \quad d \sim y_j, \quad c_j \sim x_j, \quad c_j \sim y_j. \tag{1}$$

Call the resulting system Γ.

Proposition 1. *The system Γ is a connected geometry satisfying the literal chirality definition of [1]. Its full type-preserving group is C_2. It is not firm. Moreover, all residues of corank at least two at the chamber $C_1 = \{c_1, x_1, y_1\}$ are connected, but the residue of d is disconnected.*

Proof. The only chambers are

$$C_j = \{c_j, x_j, y_j\}, \qquad D_j = \{d, x_j, y_j\} \quad (j = 1, 2).$$

Every pair in (1) lies in one of these chambers, and every object lies in a pair. Hence every flag extends to a chamber. The whole incidence graph is connected: d is joined to every x_j, y_j, and c_j is joined to x_j.

Among the type-zero objects, d has degree four and the other two have degree two. Every type-preserving automorphism fixes d. The matching $x_j \sim y_j$ forces the same permutation of the two indices on types one and two. The incidences of c_j then force that same permutation on c_1, c_2. Conversely, simultaneous exchange of the two indices preserves all incidences. Thus the full group is exactly C_2, with chamber orbits $\{C_1, C_2\}$ and $\{D_1, D_2\}$. The only adjacent chamber pairs are C_j, D_j, so adjacency always crosses between the two orbits. This is the stated chirality condition.

The panel $\{c_j, x_j\}$ has only the completion y_j, so the geometry is not firm. In fact eight panels have one completion and two panels have two completions.

The flags in C_1 of corank at least two are the empty flag and the three singleton flags. The empty residue is the connected whole geometry. The residue of c_1 is the single edge $x_1 \sim y_1$. The residue of x_1 has vertices c_1, d, y_1 and edges $c_1 \sim y_1 \sim d$; the residue of y_1 is the analogous path. All four residues are connected. In contrast, the residue of d has vertices x_1, x_2, y_1, y_2 and consists of the two disjoint edges $x_1 \sim y_1$ and $x_2 \sim y_2$. □

The example refutes three literal auxiliary assertions: the inference from chirality to thinness on printed page 493; Theorem 4.1 on page 494, applied to C_1; and Proposition 4.2 on page 495. For the last assertion, already the proper flag type $\{0\}$ has two object orbits, $\{d\}$ and $\{c_1, c_2\}$. The source proofs use an adjacent chamber where firmness has not been supplied. With firmness as an additional hypothesis, both normalization arguments are valid and are retained in the project proof bank.

This example is not a standard C^+ coset geometry. Indeed no subgroup of its full type-preserving group is transitive on the type-zero objects, whereas a coset action is transitive on each type. Thus it does not refute the corrected higher-rank variant of Corollary 4.9. The separate cyclic example handles the unrestricted wording of that corollary in rank two.

2 The rank of the residue in Proposition 4.7

The displayed statement of Proposition 4.7, printed page 496, requires $|J| < |I|$ and asserts connectedness of the residue under

$$G_J = \langle G_{J\cup\{i\}} : i \in I \setminus J \rangle. \tag{2}$$

The preceding discussion excludes rank-one residues as irrelevant to residual connectedness. This contextual restriction must also be present in the proposition: (2) does not imply connectedness in rank one. For example, in the already accepted rank-four C^+ geometry of [2], take any base panel with type set J, so $|J| = 3$. The intersection property gives $G_J = G_I = 1$, and hence (2) holds. But the residue consists of two objects of the same type. Its incidence graph has two isolated vertices and is disconnected. The existing recovery of this sufficient criterion is explicitly for residues of rank at least two and for the designated action with proper-flag transitivity. The source's separate switch between the full automorphism group and the designated group is documented in the accepted principal counterexample; neither qualification is silently removed here.

3 Verification and scope

The accompanying finite check enumerates every flag of the seven-object system and all $3!\,2!\,2! = 24$ permutations preserving its types. It finds exactly the two automorphisms proved above, the two chamber orbits, the complete panel counts and the four connected base residues. The disconnected residue of d is checked separately. No group search, classification or automorphism software is required.

The definitions and statements belong to [1]; the seven-object construction and proof are supplied here without a priority claim. The rank-one observation imports the reviewed data of [2]. These auxiliary corrections add no original, open-question solution, Lean theorem or compression measurement. The completion ledger keeps the literal assertions separate from their corrected variants.

References

[1] D. Leemans and P. Tranchida, *On residual connectedness in chiral geometries*, Algebraic Combinatorics **4** (2021), 491–499, doi:10.5802/alco.162.

[2] Human Compression project, *A chiral coset geometry with disconnected residues*, reviewed project manuscript, five pages; exact proof and review bound in the dependency ledger.